

Inflammatory optic nerve phenotypes

Part 2 | Phenotype before the antibody label

Distribution and recovery help frame the differential.

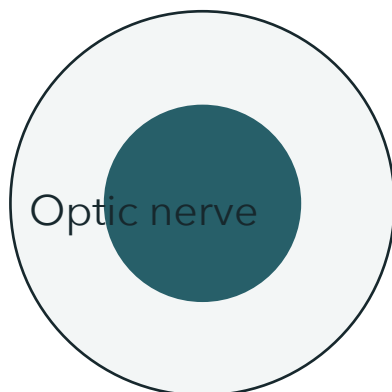
Finding	Implication
Typical MS-associated ON	Often unilateral painful loss with characteristic recovery
MOG-associated ON	Disc swelling, bilateral or anterior long-segment involvement may occur
AQP4-associated disease	Severe loss; posterior or chiasmal involvement raises concern
Atypical features	Broaden inflammatory, infective and infiltrative investigation

These are tendencies, not diagnostic absolutes. Combine examination, brain-orbit imaging, systemic symptoms and appropriately selected assays. State the finding that makes a typical demyelinating pathway unsafe to assume. Recovery, relapse and treatment dependence provide further evidence; none should be interpreted in isolation from the presenting phenotype.

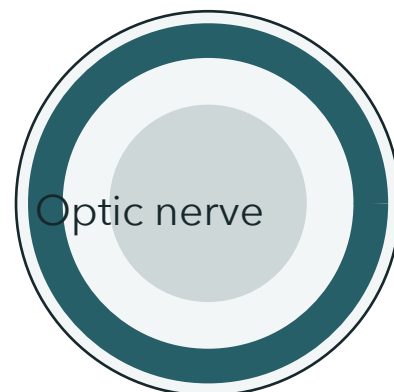
Intraneural versus perineural enhancement

Sheath enhancement is a pattern, not a unique diagnosis.

Intraneural pattern



Sheath pattern



Enhancement localisation narrows the differential; it does not uniquely establish the cause.

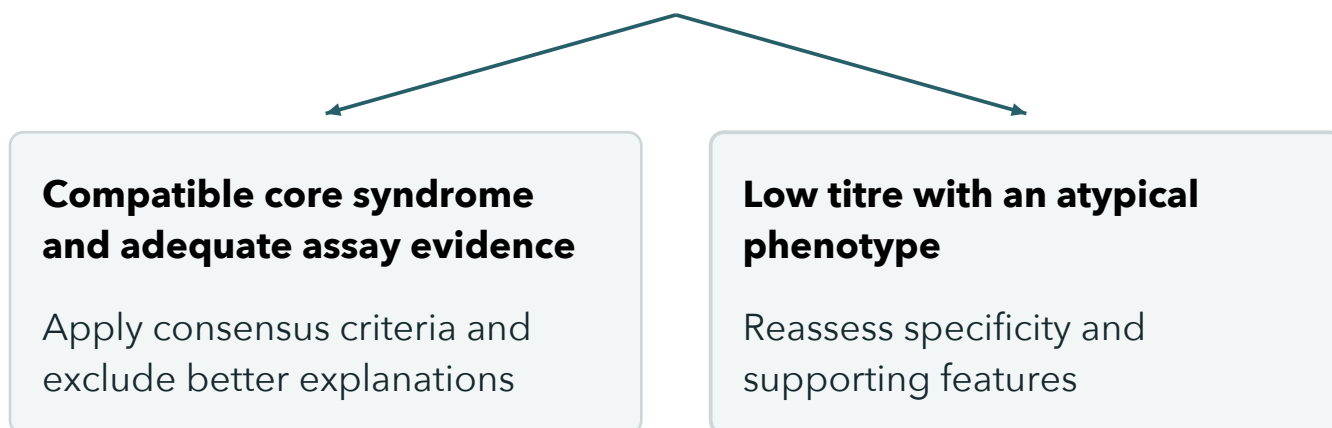
Original sections distinguish nerve and sheath enhancement compartments.

A tram-track or ring-like sheath pattern can suggest optic perineuritis, but inflammatory, infectious, granulomatous and neoplastic causes remain. Review adjacent orbital tissue and the clinical course. A presumed inflammatory diagnosis should be reconsidered when progression, systemic findings or an atypical treatment response suggest an alternative requiring targeted investigation or tissue assessment.

MOG-IgG: use pretest probability

Low-positive results need supporting clinical features.

Does a positive assay establish MOGAD?



Live cell-based testing and the clinical context matter. The positive predictive value falls when testing low-risk phenotypes indiscriminately, especially with low titres. Do not relabel an established alternative disorder on an isolated result. Explain why a confirmatory review may change long-term immunotherapy, prognosis and the interpretation of future attacks.

Acute severe loss and competing causes

Escalation should follow the actual disease and response.

- Document functional severity and imaging distribution
↓
- Exclude relevant infection and mimics
↓
- Coordinate acute treatment with neurology
↓
- Reassess response and need for specialist rescue

Severe or bilateral inflammatory optic neuropathy needs coordinated treatment and a plan for reassessment. A poor early response can make rescue therapy relevant, but the choice depends on disease, severity and competing risks. Avoid transferring the prognosis or treatment expectation of ordinary MS-associated ON to AQP4 or MOG disease.

Relapse prevention: distinguish cohort evidence from proof

Repeated attacks or steroid dependence change the maintenance discussion.

Finding	Implication
Relapsing MOGAD	Discuss maintenance treatment with neuroimmunology
IVIG evidence	Adult multicentre cohort suggests fewer relapses
Study limitation	Retrospective association, not randomised superiority
Monophasic course	Do not assume every patient needs indefinite treatment

Describe attack frequency, disability, treatment dependence and competing harms before selecting a maintenance approach. A useful evidence answer states why treatment is considered and how strongly the data support it. The reported IVIG experience should not be converted into an unqualified claim of a universally optimal regimen.